SmartDorm

Bagdat Alua

Backend (Core Logic & Auth + Integration):

Backend on Django;
JWT (login / token);
CBV (list of rooms);
Custom Manager (booking);
Postman collection;
UI: Canteen, Equipment;

Beisen Ayaulym

Backend (Models, API & Validation):

Models: Room, Equipment, Booking, IssueReport;
API: 2 CBV + 2 FBV;
Serializers (Model + regular);
Validation (occupied / date);
Permissions (IsAuthenticated);
UI: Rooms, Booking, Support;

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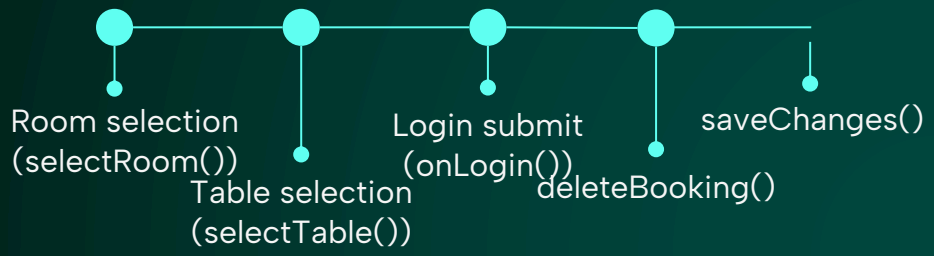
Frontend (Angular Structure & UI Logic):

Frontend on Angular;
Components and routing;
API Service + AuthInterceptor;
Booking form (ngModel, click);
Login/logout + error handling;
UI: Dashboard, My Reports;

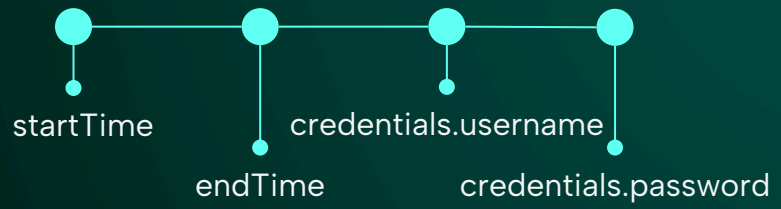
Frontend Requirements

Events & controls:

5 click events:



4 form controls:



Architecture:

- **Navigation:** 4 named routes (/rooms, /canteen, /login, /my-bookings);
- **Directives:** @if for showing notifications, @for for rendering room/table grids;
- **Auth:** JWT Authentication (stored in local storage);
- **Service:** ApiService using HttpClient for all GET/POST requests;
- **Errors:** Centralized showNotification function for all HTTP error catches

Backend requirements

Models & Logic

- 5 models:
  - Room,Equipment, Table, Booking, IssueReport
- Relationships: ForeignKey
  - ForeignKey (Booking to user)
- Manager: Custom BookingManager
  - Custom BookingManager

API and views

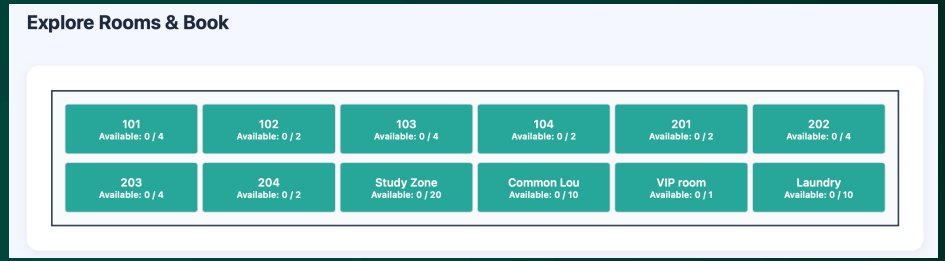
- **Serializers:** 2 serializers. Serializers; 2 serializers. ModelSerializer
- Views: FBV for simple status chek;(cancel_booking,create_issue) CBV for CRUD operations(RoomListView, BookingCreateView)
- Auth: Token-based authentication
- CORS: Configured via django-cors-headers

VP of Operations

- 4 CRUD Ops
  - Create Booking, Read Tables, Update Table Status, Delete Booking
- User Link
  - Each booking is automatically linked to request.user

Project Summary

Our project is a unified dashboard for campus resource management. It allows students to check real-time availability and book study rooms or canteen tables through a single interface.



Uniqueness:

The system features a dynamic visual map for both rooms and canteen tables, providing instant visual feedback on occupancy status.